QC STORY PROJECT

To Eliminate Scrap of Back Door Inner Assly of Scorpio BIW.

Mahindra and Mahindra Ltd, Dept : Scorpio Body Shop, Plant : Nasik

	Plan	Actual
Start Date	**9/7/2012**	**6/8/2012**
Completion Date	**12/10/2012**	**02/11/2012**

Current Status:

Completed.

Cross Functional Team

No	Description	Name	Designation	Dept	Phone	Email
1	**Team Leader**	**CHAITANYA DASAKKAR**	ASST. MANAGER	SCORPIO BODYSHOP	8605012840	DASAKKAR.CHAITANYA2@ mahindra.com
2	**Mentor**	**SURYAVANSHI R G**	DGM & HOD SCORPIO BODYSHOP	SCORPIO BODYSHOP	9881378988	SURYAVANSHI. RAVINDRA@ mahindra.com
3	**Member 1**	**MANOJ MORE**	DEPUTY. MANAGER PRODUCTION	SCORPIO BODYSHOP	9922644399	MORE.MANOJ@ mahindra.com
4	**Member 2**	**BHAT VIVEK**	DEPUTY MANAGER TOOLING	SCORPIO BODYSHOP	9922438403	BHAT.VIVEK@ mahindra.com
5	**Member 3**	**AMRUTKAR GOKUL**	DEPUTY MANAGER QUALITY	SCORPIO BODYSHOP	8888814357	AMRUTKAR.GOKUL@ mahindra.com

1. Select the Theme

Brief Description of Problem

What: **Repetitive Heavy Damage to Back door inner Panel asslys.**

Where: **Back Door Assly station No-10.**

When : **Daily (1st & 2nd shift)**

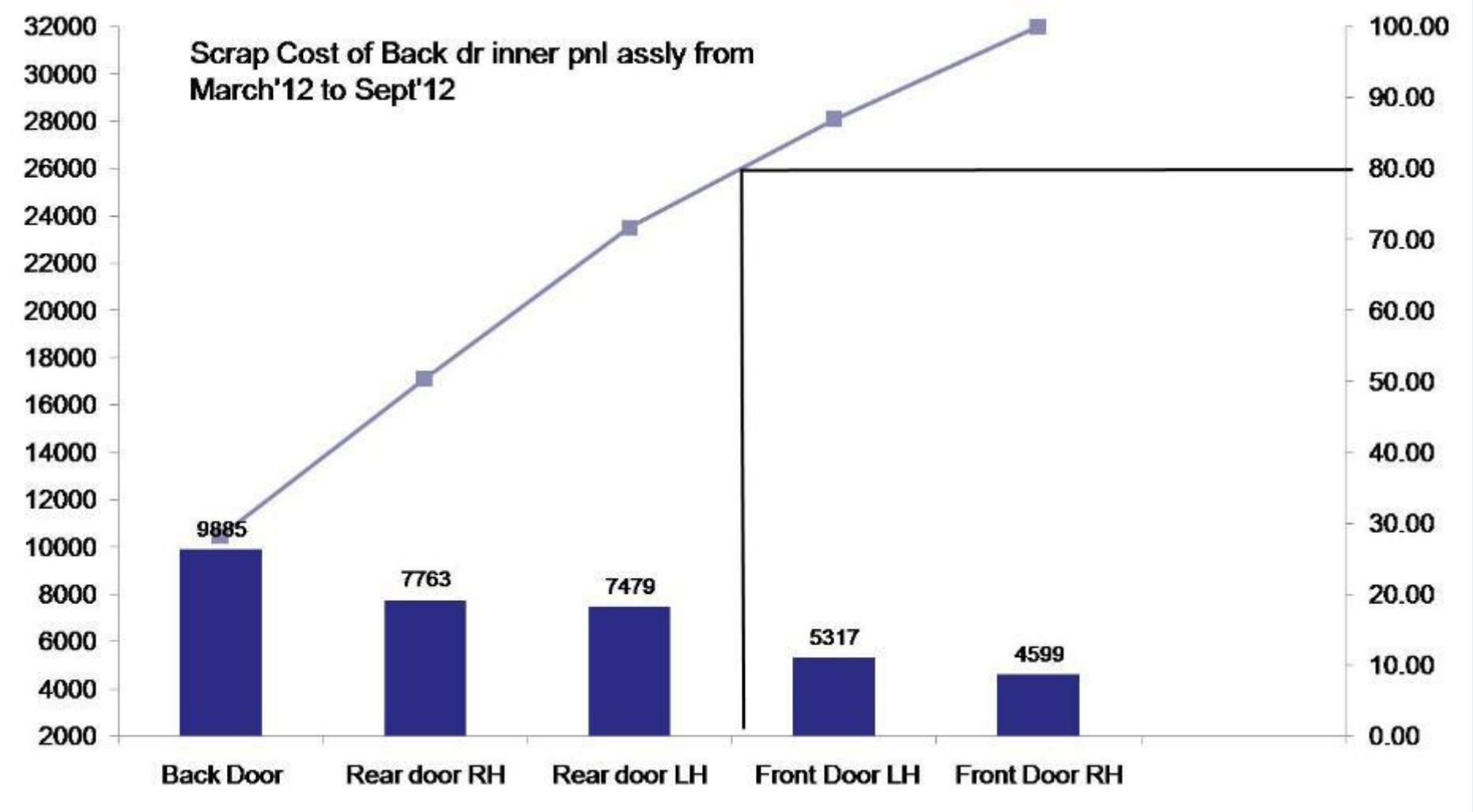
Who : **Concern reported by Stage operator, Buyoff operator, check man.**

Why : **High Scrap Cost of door & Cost Of Poor Quality.**

How: **Average Incidences 3 per week having V1 severity**

2. Justify the choice (reasons)

Scrap Cost of Back Door Inner panel assly from March'12 to Sept'12



2. Justify the choice (reasons)

INFERENCE

As Back Door Scrap Cost , Rework Time & COPQ is high, Concern taken for solving through QC Story

2. Justify the choice (reasons)

Mahindra Automotive Sector

PROMISE 2013

- We will be amongst the Top 3 in customer satisfaction in India.
- We will be a $5bn (Rs. 23,000 cr) business.
- We will sell more utility vehicles in India than all our competitors combined. We will be India's 2nd largest commercial vehicle manufacturer.
- We will achieve an overseas volume of 65,000 vehicles, with a successful US presence.
- We will establish MRV and launch 10 new products with multiple variants.
- We will demonstrate our commitment to sustainability with new hybrid and electric vehicles.
- We will continue to focus on successful business performance and achieve double digit OPM by creating products with differentiated value propositions.

We will enable our people and customers to RISE as we embark on our journey towards being a Top 10 global automotive brand.

Nashik Plant

Mahindra Automotive Sector

PROMISE 2013

- Achieve Double Digit 3 MIS for Build Quality of Nasik products.
- We shall ensure Zero PDI defects at Dealer @ zero KM.
- We will produce 200,000 vehicles in a year.
- We will support marketing to meet the customer demand by delivering 100% vehicles as per e-DCM committed dates.
- We will be ready to produce 50,000 Export vehicles.
- We will ensure ramp up of new products as per Launch Plan.
- We will strive for reducing consumption of natural resources used in our manufacturing process.
- Embracing Innovative Ways and New Technologies to continually reduce the Operational Cost by 25% over F10.

We shall strive to achieve Level 6 in MQW

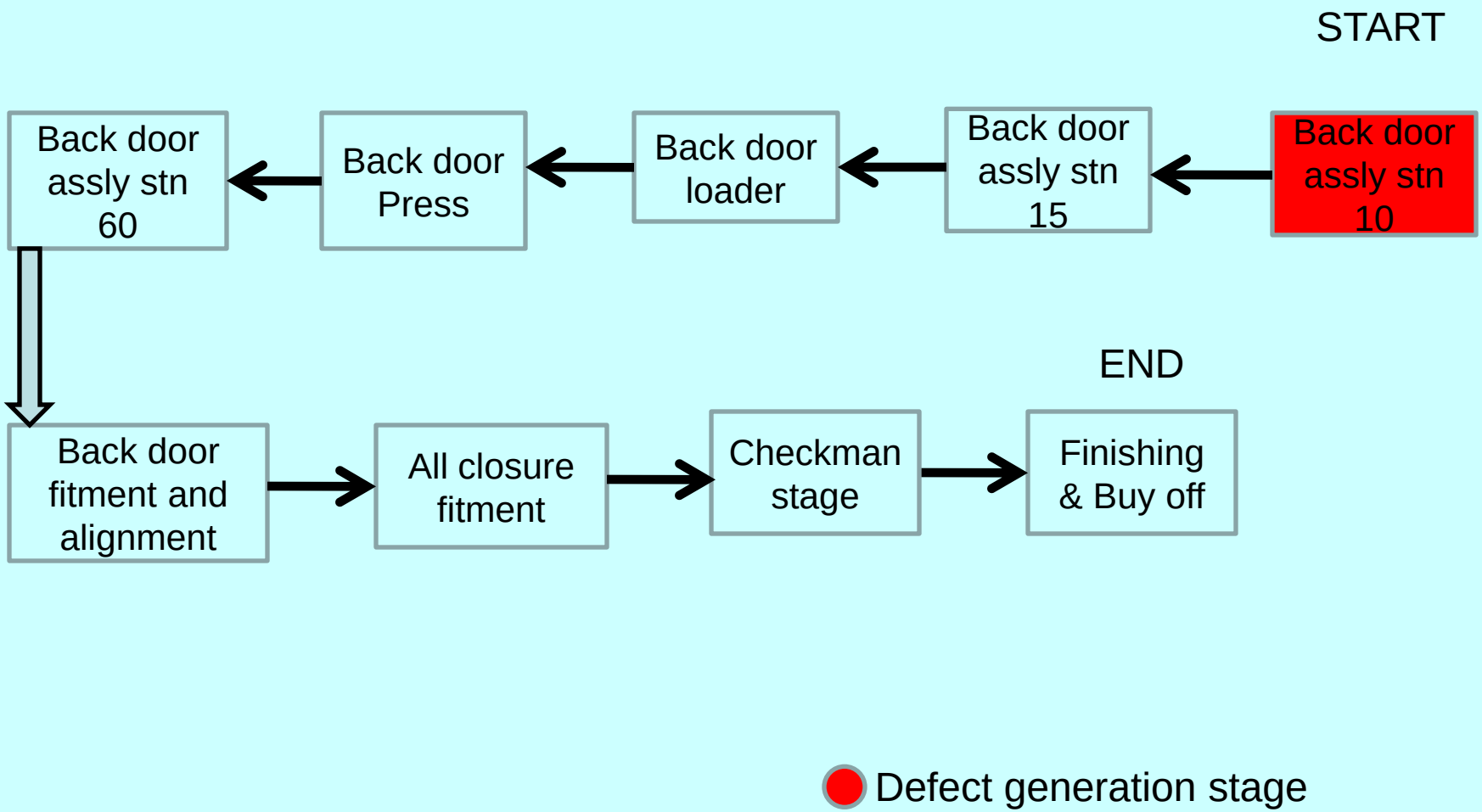
2.a Immediate Containment Action

Emergency Response

- 100% inspection at Back Door Stage.
- 100% inspection at Buy-off Stage.

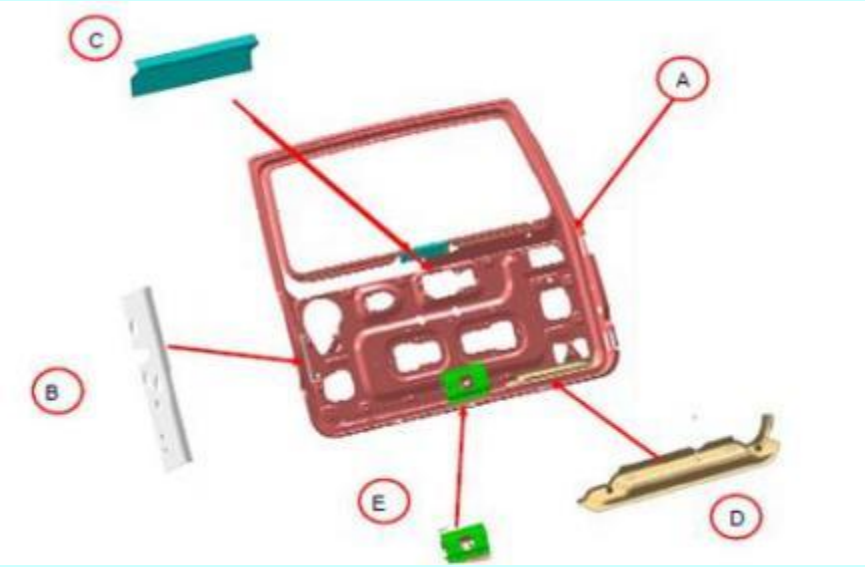
3. Understand the current situation

Flow Chart



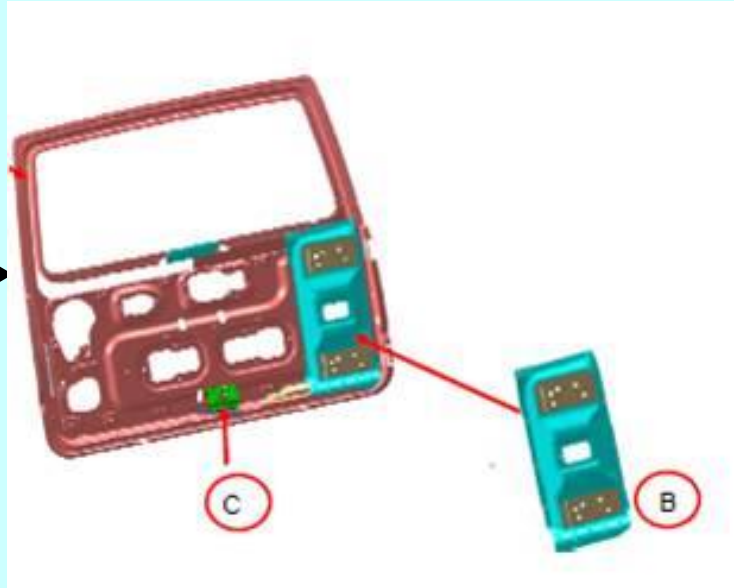
PROCESS FLOW CHART

BACK DOOR ASSLY STN-10



Parts assembled to Back Door inr Panel
1.Check arm mtg brkt
2.Latch reinforcment
3.Grab handle mtg brkt.
4.Fog lamp mtg brkt

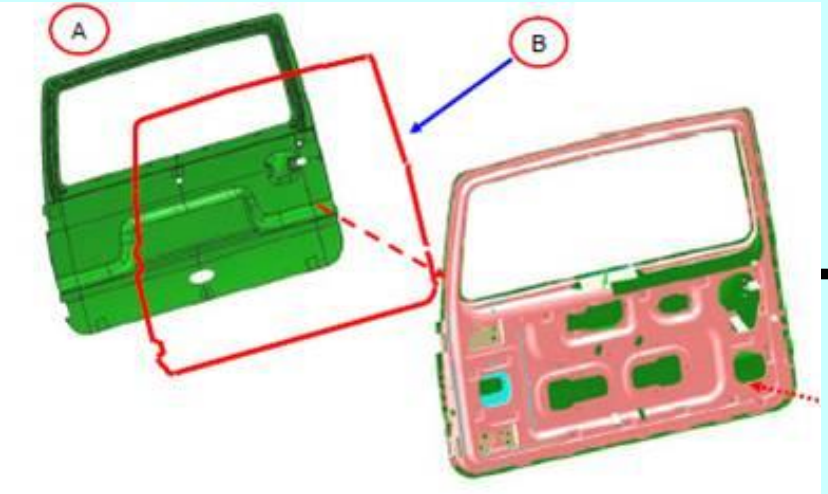
BACK DOOR ASSLY STN-15



Parts assembled to Back Door inr Panel
1.Hinge mtg brkt
2.Mastic sealer
3.Fog lamp mtg brkt

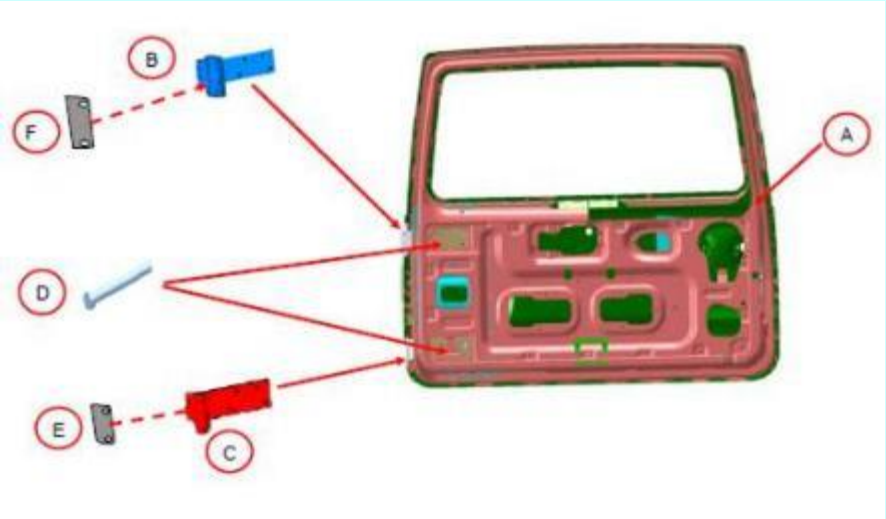
PROCESS FLOW CHART

BACK DOOR ASSLY STN-20



Parts assembled to Back Door inr Panel
1. Back Door inr pnl

BACK DOOR ASSLY STN-60



Parts assembled to Back Door inr Panel
1. Hinge top
2. Hinge bottom

PROCESS FLOW CHART

BACK DOOR ASSLY STN-10

INR PNL TO LATCH ASSLY

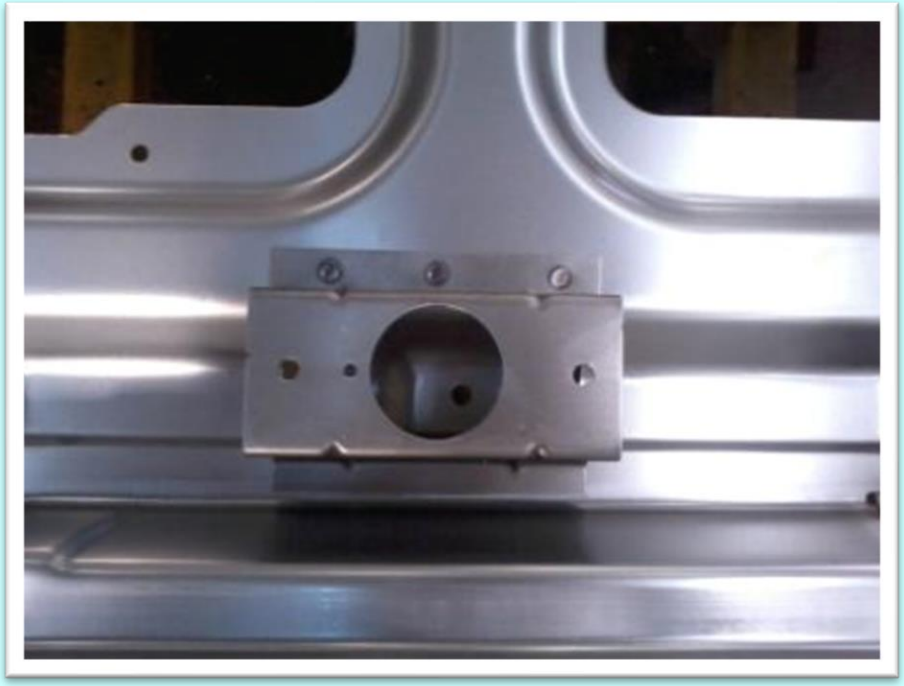
INR PNL TO GRAB HANDLE ASSLY

START

PROCESS FLOW CHART

BACK DOOR ASSLY STN-10

INR PNL TO FOG LAMP ASSLY



INR PNL TO CHECK ARM MTG ASSLY

→

END

3. Understand the current situation

- Heavy damage of Back door due to lifter at stn-10

3. Understand the current situation

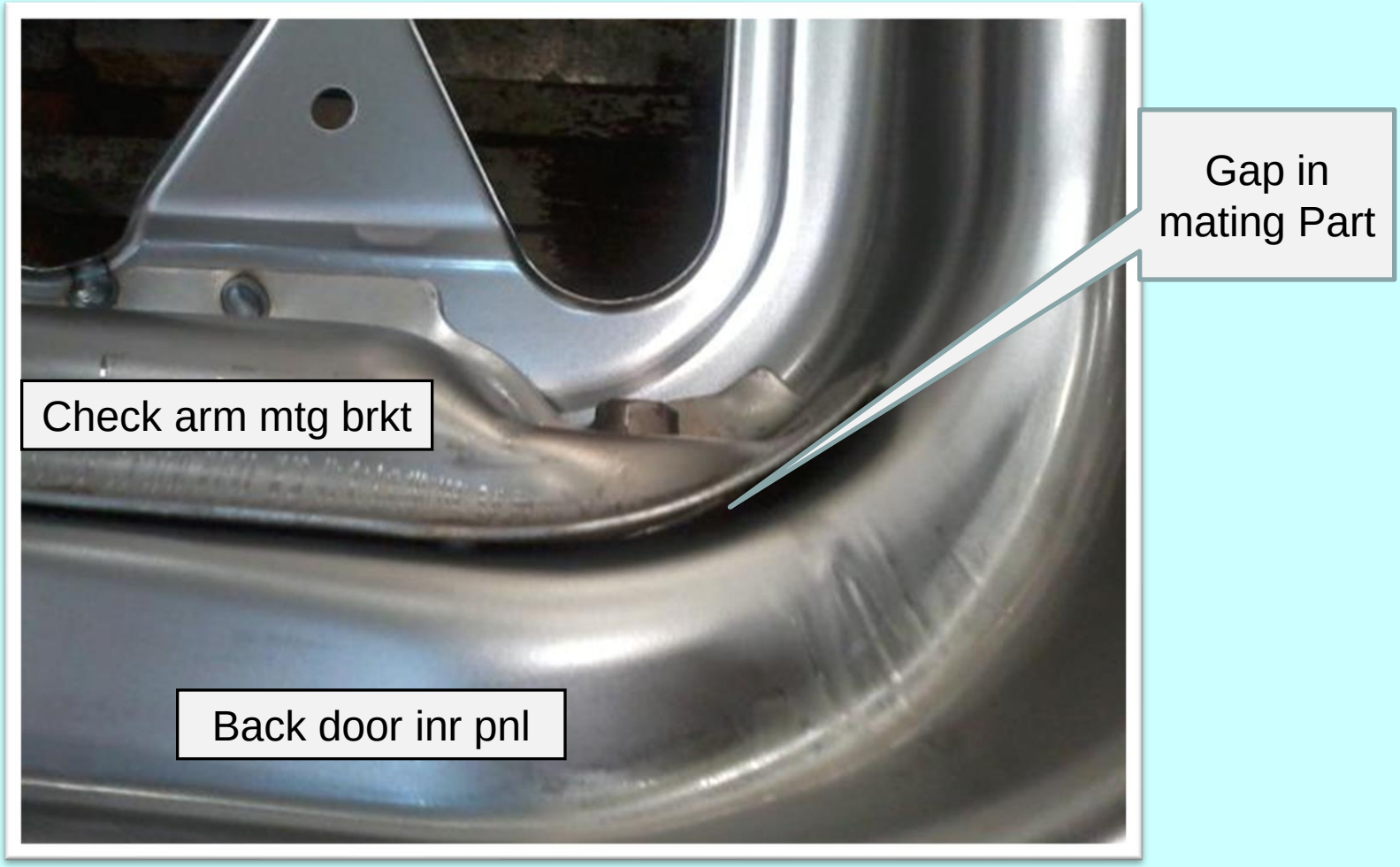
- Latch assly damage at stn-10

3. Understand the current situation

- Spot hole at check arm mtg brkt at stn- 10

3. Understand the current situation

- Gap in mating part



3. Understand the current situation

- Check arm flange Spot hole

3. Understand the current situation

- Grab handle area damage

3. Understand the current situation

- Cost of Poor Quality is High due to Back Door Assly Scrap/Rework.
- Gap in Check arm Mounting Bracket Flange & Back door inner pnl Assy.
- Latch Reinf having Panel mismatch while locating over panel.

4. Select Target

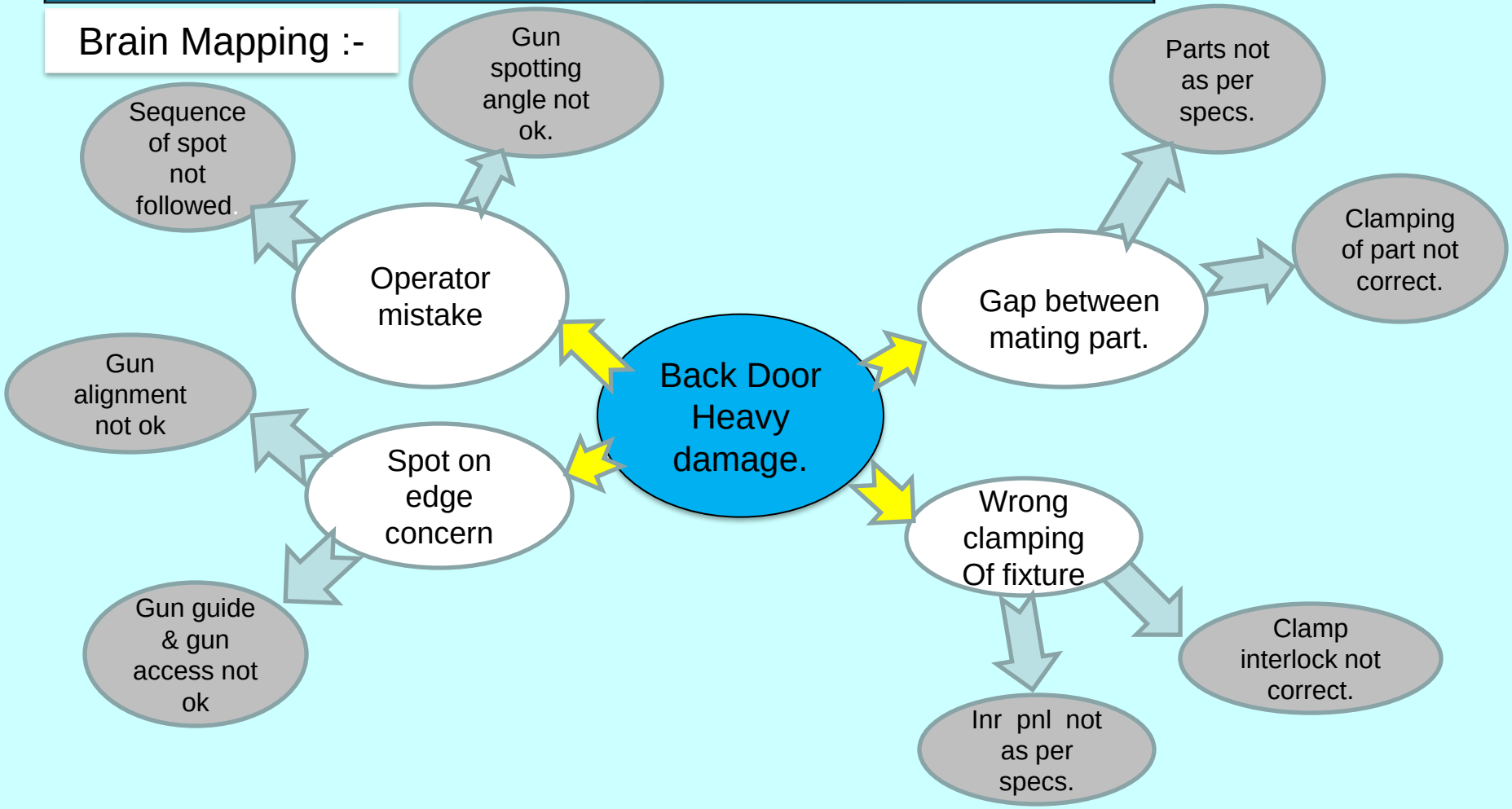
Identify the target. A target is a number indicating the level of improvement to be achieved.

Present Level:	Scrap Cost per Month Rs 1412
Target:	Scrap Cost per Month is Zero.

5. Analysis

5.1 Identify possible causes:

Brain Mapping :-



5. Analysis

5.1 Identify possible causes:

1. Gap between the both the panel.

2. Panels are mismatch.

3.Spot on edge concern.

4. Spot welding parameters not ok

5. Gun alignment not ok

6. Frequent tip worn out

7. Excessive heating of tip

8.Gun angle not ok for spot welding

9. Clamping sequence not correct.

5. Analysis

5.1 Identify possible causes:

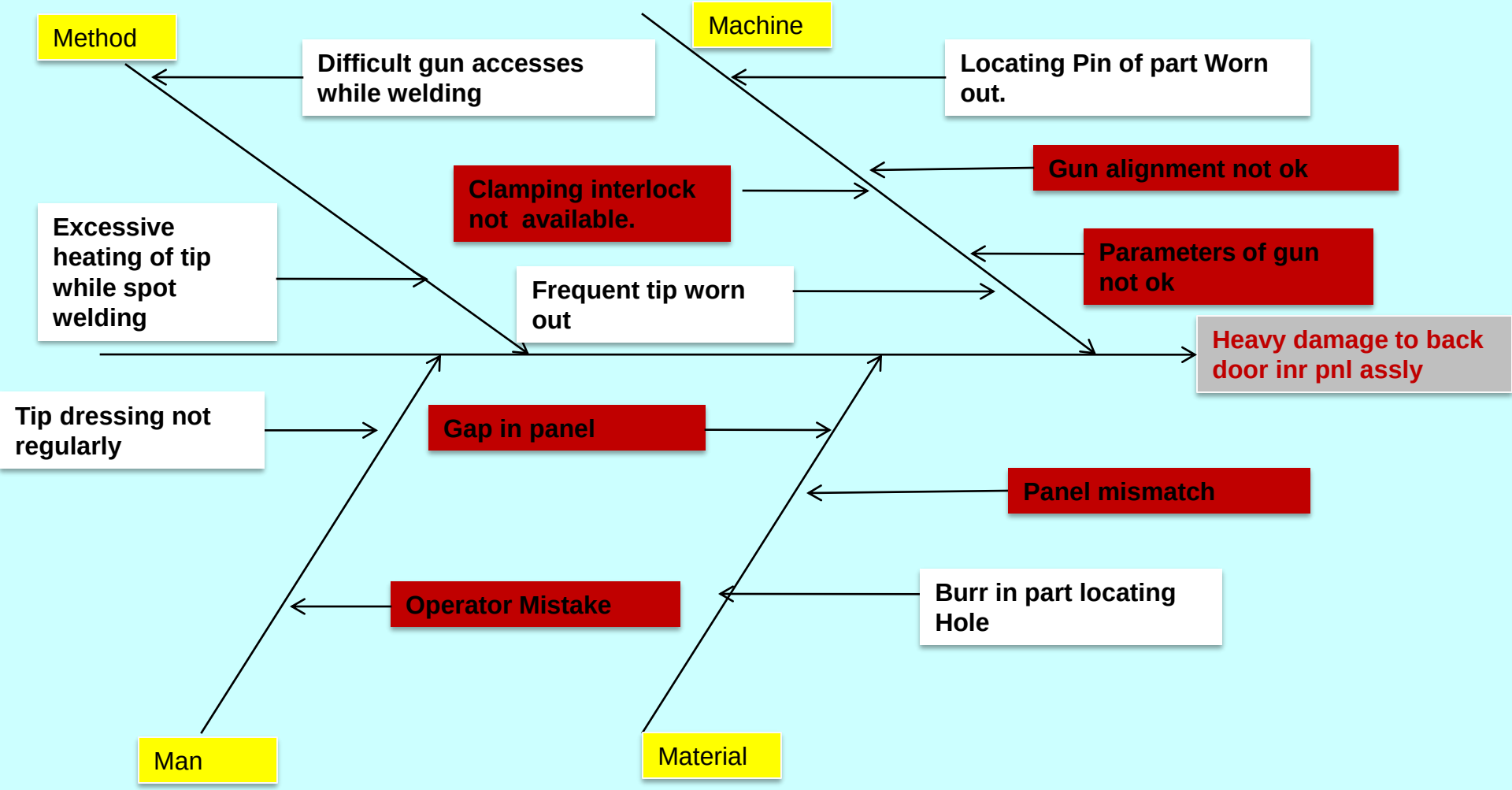
10. Operator Mistake.

11. Burr in Part Locating Hole.

12. Welding Gun access not Ok for all parts.

5. Analysis

5.1 Identify possible causes:



Note- Red blocks indicates the Probable causes

5. Analysis

5.2 List of Probable causes:

1. Gap between the panel.

2. Panel mismatch.

3. Welding Gun Parameters not ok.

4. Gun alignment not ok.

5. Damage due to wrong clamping.

6. Damage due to Operator Mistake.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing Method	Observations	Conclusion
1	Gap between Check arm mtg brkt & inner pnl.	By Vernier caliper	100 parts out of 100 found having gap of 1.2-2.0 mm against complete matching of parts	**Hypothesis valid**

Some readings are shown

Sr No.	Gap(in mm)	Sr No.	Gap(in mm)
1	1.5	11	1.8
2	1.6	12	1.6
3	2	13	1.6
4	1.1	14	1.9
5	1.7	15	1.1
6	1.6	16	1.8
7	1.2	17	1.9
8	1.8	18	1.6
9	2.0	19	1.5
10	1.7	20	1.6

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing Method	Observations	Conclusion
2	Panel mismatch between Latch reinforcement & back door inner panel.	Visual, No mismatch	64 sides out of 100 found having panel mismatch	**Hypothesis valid**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing Method	Observations	Conclusion
3	Welding Gun parameters not ok	Refer design parameter	Parameters found ok with respect to design parameter of both guns	**Hypothesis invalid**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing Method	Observations	Conclusion
4	Gun alignment not ok	**Visual**	Gun alignment of shank found ok.	**Hypothesis invalid**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing Method	Observations	Conclusion
5	Damage due to wrong clamping	**Production trail taken on fixture**	Wrong clamp sequence damaging inr pnl assly	**Hypothesis valid**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing Method	Observations	Conclusion
6	Damage due Operator Mistake	**Production trail taken on fixture with operator.**	Wrong Gun handling & spoting leading to damage to inner pnl assly.	**Hypothesis valid**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

SR. No.	Probable Cause	Testing and Observations	Conclusion
1	Gap between Check arm mtg brkt & inner pnl.	100 parts out of 100 found having gap of 1.2-2.0 mm against complete matching of parts	**Hypothesis valid**
2	Panel mismatch between Latch reinforcement& back door inner pnl.	64 sides out of 100 found having panel mismatch	**Hypothesis valid**
3	Welding Gun parameters not ok	Parameters found ok with respect to design parameter of both guns	**Hypothesis invalid**
4	Gun alignment not ok	Gun alignment of both shank found ok	**Hypothesis invalid**
5	Damage due to wrong clamping	Wrong clamp sequence damaging inr pnl assly	**Hypothesis valid**
6	Damage due Operator Mistake	Production trail taken on fixture with operator.	**Hypothesis valid**

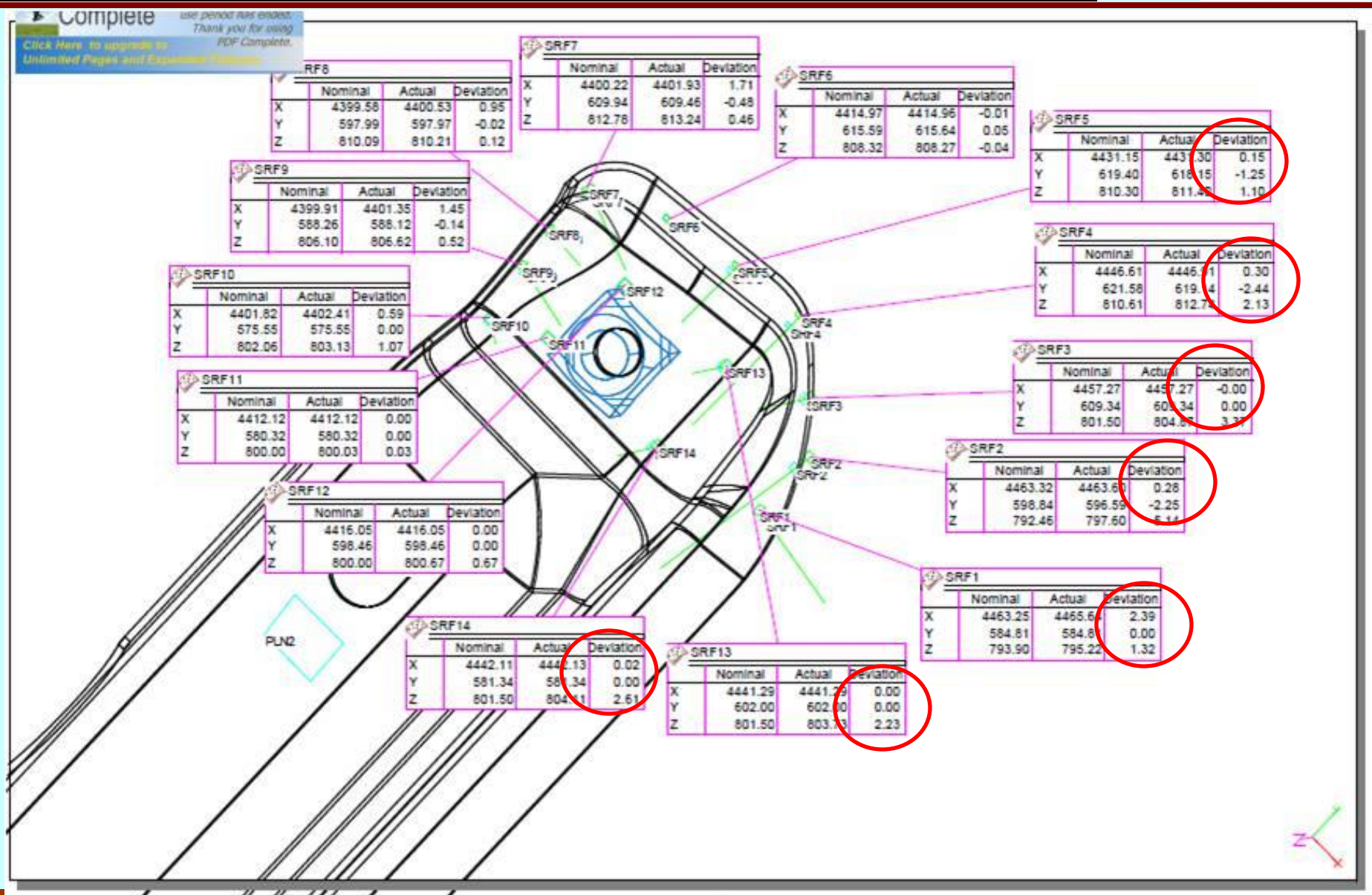
5.3 Identify the root causes: (Use Why – Why)

Sr No	Valid Probable Cause	WHY	WHY	WHY	WHY	ACTION
1	Gap between Check arm mtg brkt & inner pnl.	Bracket curved profile not mating to inr pnl	Heavy gap in Bracket profile & inner pnl..	Check arm bracket profile not as per part drawing.	Die worn out of Check arm.	**To confirm part as per part drawing.**
2	Panel mismatch between Latch reinforcement & back door inner pnl.	Latch reinf. Not located correctly over fixture.	Fixture clamp & pin having extra play.			**Sliding clamp & pin to be replaced with swinging clamp**
5	Damage due to wrong clamping	No sequencing of clamp.	Interlocks are not provided on fixture.			**Clamp interlocking to be provided.**
6	Damage due to Operator Mistake.	Gun Movment difficult.	Extra efforts req to pull gun.	No control on Gun movement for particular area of fixture.		**Gun Locating & Guiding guard to be provided .**

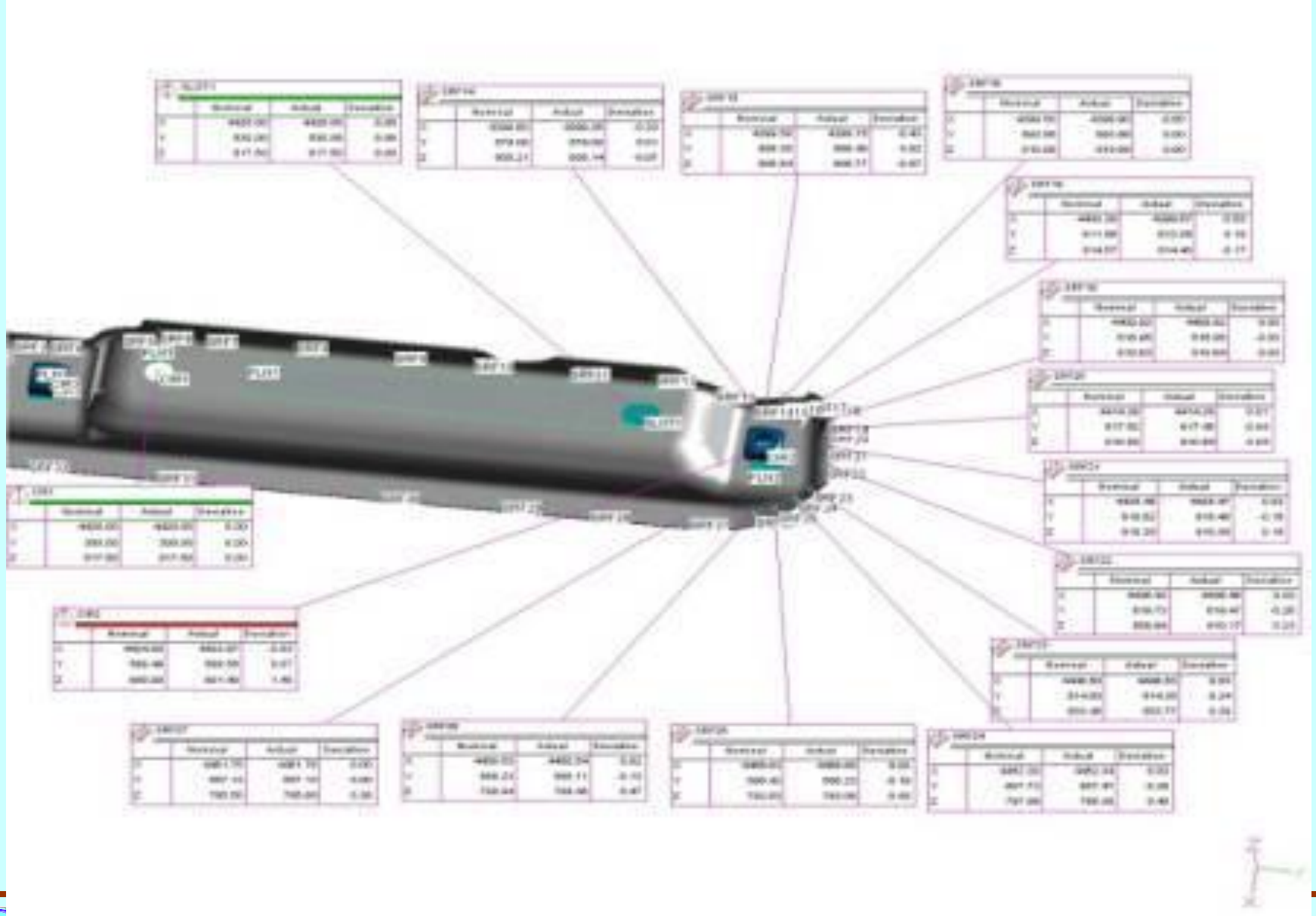
6. Implement The Corrective Measures

Root Cause	Action Plan	Target Date	Resp	Status
1. To confirm part as per part drawing	1.CMM to be done of part & part deviation to be communicated to supplier. 2.Part to be corrected as per part drg & checked by CMM.	4.10.12	Mangesh	Closed

Before: Report of Check arm part on CMM



After: Report of Check arm part on CMM



6. Implement The Corrective Measures

Root Cause	Action Plan	Target Date	Resp	Status
2.Panel mismatch between Latch reinforcement& back door inner pnl.	**Sliding clamp & pin to be replaced with swinging clamp & pin with groove to be provided on locating block for Latch V cutout..**	**25/8/2012**	**1.Vivek** **2.Chaitanya**	**Closed**

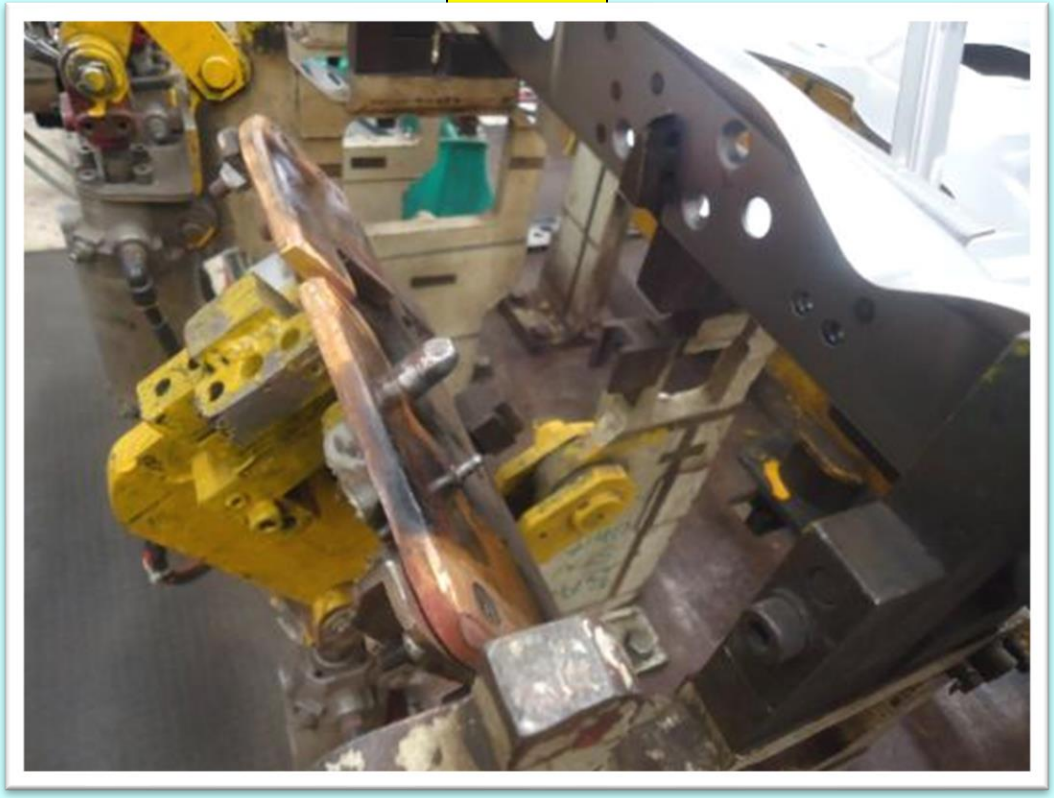
6. Implement The Corrective Measures

Before

NA

Sliding clamp Mechanism

After



Swiveling clamp with grooved block.

6. Implement The Corrective Measures

Root Cause	Action Plan	Target Date	Resp.	Status
3. Damage due to wrong clamping	Clamp interlocking to be provided.	4/9/2012	1.Vivek 2.Chaitanya	Closed

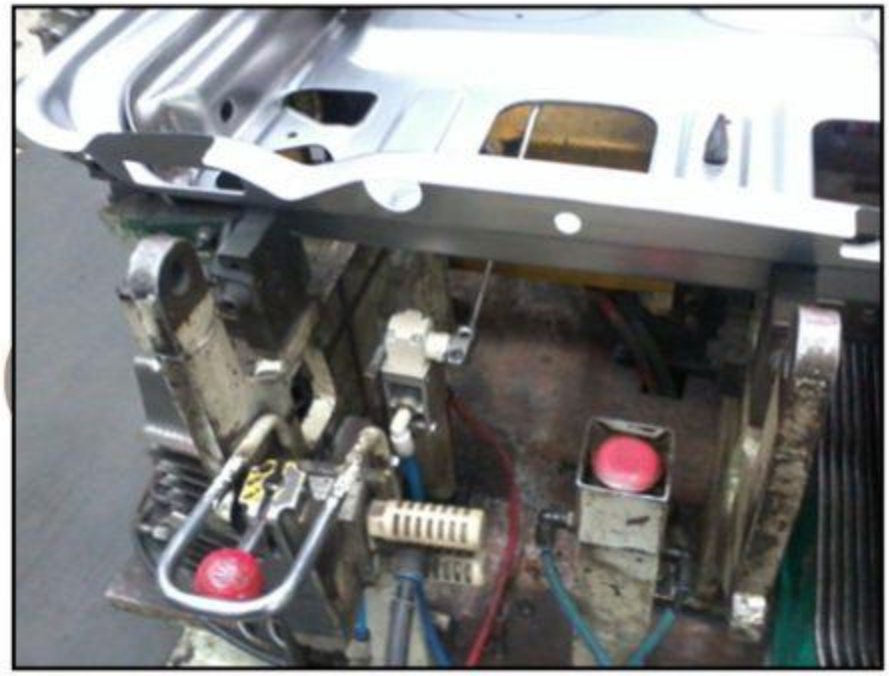
Limit switch provided at lifter so that after Inr pnl will be at BDC, then only next sequential clamp will operate.

Improvement Sheet

Date: 4/9/2012

Name:- Scorpio Body Shop **BWT / Equip. Name** :- LH Side Press/Back Door asssly#10

eme :- Reduction in Scrap Cost

Before	After
	
1 Wrong location of Back door inr pnl over fixture after lifter down.	1 Limit switch provided below Back door inr pnl.
2 Heavy damage to inner pnl after clamping door over fixture.	2 clamp interlocking done for lifter down, clamping of doo clamping of parts.

nefits achieved in PQCDSM

:- Elimination of damage of inr pnl due to wrong clamping resulting reduced scrap cost.

6. Implement The Corrective Measures

Root Cause	Action Plan	Target Date	Resp.	Status
4.Damage due to Operator Mistake.	**Gun Locating & Guiding guard to be provided . (2** Locations)	17/9/2012	1.Vivek 2.Chaitanya	Closed

Improvement Sheet

Date: 17/9/2012

PU Name:- Scorpio Body Shop **BWT / Equip. Name** :- LH Side Press/Back Door asssly#10

Theme :- Reduction in Scrap Cost

Before	After
NA	
1 No restriction to Gun Movement while taking spot on check arm flange. 2 Heavy damage & Spot hole to inner pnl & Check arm brkt after Spoting part.	1 Copper plate designed as a Guard for Spot Welding.

Benefits achieved in PQCDSM

1. C- Elimination of damage of inr pnl & Check arm due to Wrong gun handling by operator Mistake.

Improvement Sheet

Date: 17/9/2012

PU Name:- Scorpio Body Shop

BWT / Equip. Name :- LH Side Press/Back Door asssly#10

Theme :- Reduction in Scrap Cost

Before

After

1	No restriction to Gun Movement while taking spot on grab handle Mtg brkt.	1	Nylon Guard is designed for restriction of gun into inr pnl Profile.
2	Heavy damage & Spot hole to inner pnl & Grab Handle brkt after Spoting part.		

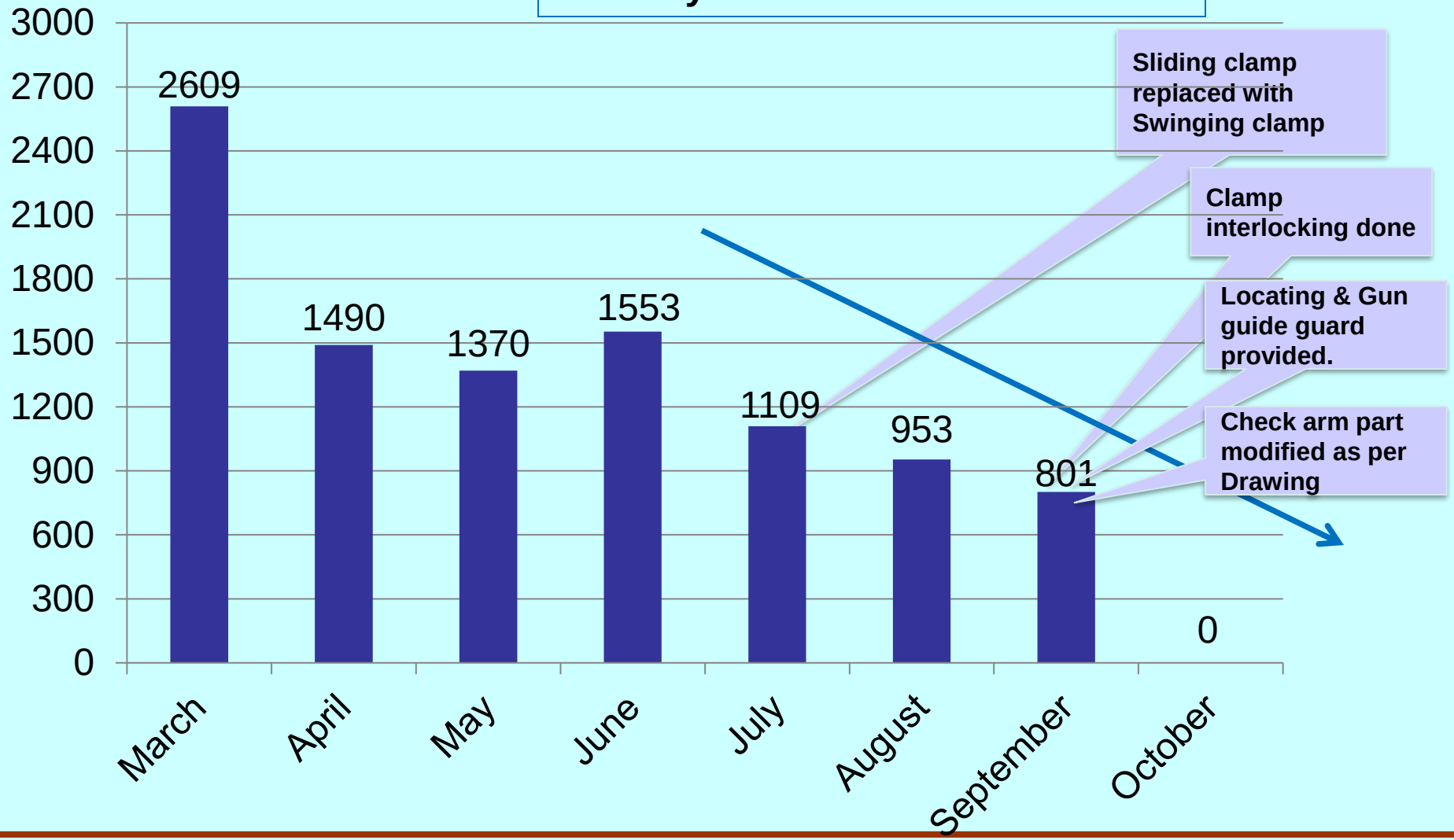
Benefits achieved in PQCDSM

1. C- Elimination of damage of inr pnl & Grab handle due to Wrong gun handling by operator Mistake.

7. Check & Confirm the Results & Effects

7.1 Verify Results:

Data of Scrap Cost of Back Door Inr assly Per Month



7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

1.All decided actions implemented as on 4/10/2012

2. Effectiveness of actions has been audited on 4/10/2012 by MQA team

3. Frequent Supplier audit by CFT.

7. Check & confirm the results & effects (Cont…)

7.3 Tangible

1.Back door inner panel assly Scrap Cost reduced from Avg monthly Rs 1412 to Zero.

2. Annual Cost saving of Rs16944.

3. Zero incidences at TCF,CAI,RFI,PDI from last 1 months.

4. COPQ for April to Sep'12 brought down compared to April to Sep '11 for Back door heavy damage concern.

7. Check & confirm the results & effects (Cont…)

7.3 Intangible benefits:

1. Morale improved
2. Team Work Builtup
3. Confidence Boost up

8. Standardisation

8.1 Updation of documents:

1. Fixture History card updation

FIXTURE HISTORY CARD					
ZONE:	Z20				DEPT : MFG ENGG.-SCORPIO BODY
FIXTURE NO.	Z20-10		FREQUENCY : 6 Months and/or whenever changes done		
ASSY NAME:	Assy Back door 10				
DATE	UNIT NO	ITEM NO.	CONCERNS	CHANGES DONE/ ACTION TAKEN	REMARKS
15/10/09			Damage near grab handle	MS Plate provided to reinforce nylon guard.	i4-107854
12/11/2009			Damage foglamp mtg brkt	Magnet provided on locating pin	Scrap Nov-09
20/11/09			Heavy spot damage on latch area	Engraving provided on copper plate	Scrap\fbo concern in Nov-09
30/1/10			Latch mtg screw hole oversize	Additional plate welded to v block.	CIA (CA3P44/09)
1/10/2010			Damage check arm flange	check arm flange guide extended	QCRT CA3P0075/10
6/11/2010			Back Door Latch Area Spot Damage	Guard provided to avoid spot on cu plate edge	
21/12/10			Damage back door Grab Handle	Cu plate float restriction by providing support to the plate opposite to the Pin	QCRT CA3P0081/10
15/1/11			Dent-Inner panel	Gap created in panel and myler in asthetic area by removing shims and providing additional resting at the flange.	
18/05/11			Heavy mismatch latch area	Additional stopper provided to arrest the latch brkt movement	Scrap-May-11
10/7/2011			Heavy damage on latch hole	Copper plug provided	Scrap July-11
24/07/2011			Back door latch area pin hole	New copper plate provided	CA2P0416
16/12/2011			Damage check arm mtg area	Stopper provided	CA2P0495
16/01/12			Spot twist on latch area	Stopper provided to avoid spot on edge	CA2P00508
24/1/12			Back door damage near Grab handle	Profile cu plate provided of Grab handle	i4-68829
16/03/12			Damage back door latch area	V block groove \relief provided	CA2P0538
25/3/12			Damage in Check arm area inside back door	Provided interlocking in Check arm clamp	i4-64481
24/4/12			Damage at check area while spotting	Provided cu plate guard instead of Nylon	i4-85474
20/5/12			Spot damage upper grab handle	Provided fillet to profile shape cu plate at grab handle	
20/5/12			Damage at check area while spotting	Nylon guard provided for check arm Bkt	i4-91539
29.05.2012			Inner lower portion of check arm spot hole\Damage	Provided cu plate guard	CA3P0144
25/8/2012			Latch reinf milmatch with inner panel	Sliding clamp replaced with swing type clamp	Chronic concern action
4/9/2012			Inner panel damage, due to wrong clamping sequence.	Interlock provided between clamps and lifter	Chronic concern action
17/9/2012			Inner panel & Check arm area Spot hole.	Copper guard provided for Gun restriction	Chronic concern action
17/9/2012			Inner panel & Grab Handle area Heavy damage.	Nylon Guard provided for Gun guiding & Stopper purpose	Chronic concern action

Page 1

8.1 Updation of documents:

2. Tool PM History card

VRM METAZINE PVT LTD W-189, MIDC, Ambad, Nashik 42 ISO/TS 16949	TOOL PM HISTORY CARD Plant: VMPL	Tool No : 0103CG0210N-20 Tool Desc : FORMING TOOL

PM Freq (Qty): 10000 Tool Life (Qty): 500000 Strokes Tool Life Start Date : 31/01/2012

PM Start Date	PM No	PM at Prod Qty	Major work done / Action taken	Done by	Approved by
05/09/2012	1	10140	New punch and die fitted in forming tool. Spotting,polishing,stopper setting done, part CMM done and found OK. The cost of subject modification is Rs.60000/-	Deepyog Engg	SACHIN YADAV

Printed by SACHIN YADAV on 31/10/2012 12:25:39 PM

Tool PM History Card

8.1 Updation of documents:

3. Tool PM History card

VRM METAZINE PVT LTD W-189, MIDC, Ambad, Nashik 422 0	TOOL PM REPORT Plant : VMPL	ISO/TS 16949 PM No : 1

Tool No :	0103CG0210N-20	ToolName :	FORMING TOOL	PM Freq / Done Qty :	10000 / 10140
Part No :	0103CG0210N	PartName :	BRKT CHECK ARM BACK DOOR	PM Start Date/Time :	25/09/2012 12:09
Opn No :	20	OpnName :	FORMING	PM Finish Date/ Time :	04/10/2012 10:00

ToolType : Plate Tool Category : Forming Freq: 270

SrNo	Checkpoints	Work Instruction	Status	Remark
1	Overall die condition	Washing / Cleaning	Done	
2	Production Feed back	Refer Prod. / Inspection / Die maint. log book	Ok	
3	Punch surface	Check for Damage marks, breakage, bulging, scoring. Add material by welding, Polish the punch, Match the profile. Do spotting with die.	Checked	New Forming Punch added to tool.
4	Cavity surface	Check for Damage marks, breakage, bulging, scoring. Add material by welding, Polish and match the profile. Do spotting with punch.	Checked	New Die Block added to tool
5	Blank Holder surface	Check for Damage marks/breakage/bulging/scoring. Add material by welding, Polish and match the profile. Do spotting with die. Clean and polish bottom surface.	Checked	Found Ok
6	Punch Inserts Fitting	Check for bolts and dowels in place and good condition. Replace the bolts and or dowels if required. Match the inserts profile by proper fastening.	Checked	Found OK
7	Die Inserts Fitting	Check for bolts and dowels in place and good condition. Replace the bolts and or dowels if required. Match the inserts profile by proper fastening.	Checked	Found Ok
8	Blank Holder Fitting	Check for bolts and dowels in place and good condition. Replace the bolts and or dowels if required. Match the inserts profile by proper fastening.	Checked	Found OK

8.1 Updation of documents:

3. Tool PM History card

9	Blank Locator	Check for damage, loose fitting, breakage.	Checked	Stopper setting done, trial taken, found OK
10	Cushion Pin Holes	Check for damage, bulging. Clear the holes with pencil grinder.	NA	
11	Pneumatic connection	Check for piping leakage	NA	
12	Fastener checking	Ensure tightness of Wear plate, Locators, Stripper, Balancer block, Bottoming block & bottoming mark bolt.	Checked	Found OK
13	Chrome plating	Check for pill-off.	NA	
14	Oiling	To all the machined surfaces in the die including top & bottom surfaces	Done	
15	Greasing	To all the wear plates / guiding surfaces / bushes & pillers etc.	Done	
16	Die Top / Bottom surface	Ensure for oiling and cleaning	Done	
17	Wear Plates / Guides	Check for Scoring / Roughness	Checked	Found OK
18	Bottoming marker	Checking for missing / Damage.	NA	
19	Air Vent Holes	Check for choking / Jammed.	Checked	Found OK

8.1 Updation of documents:

3. Tool PM Maintenance Schedule

VRM METAZINE PVT LTD
W-189, MIDC, Ambad, Nashik 422 01
ISO/TS 16949

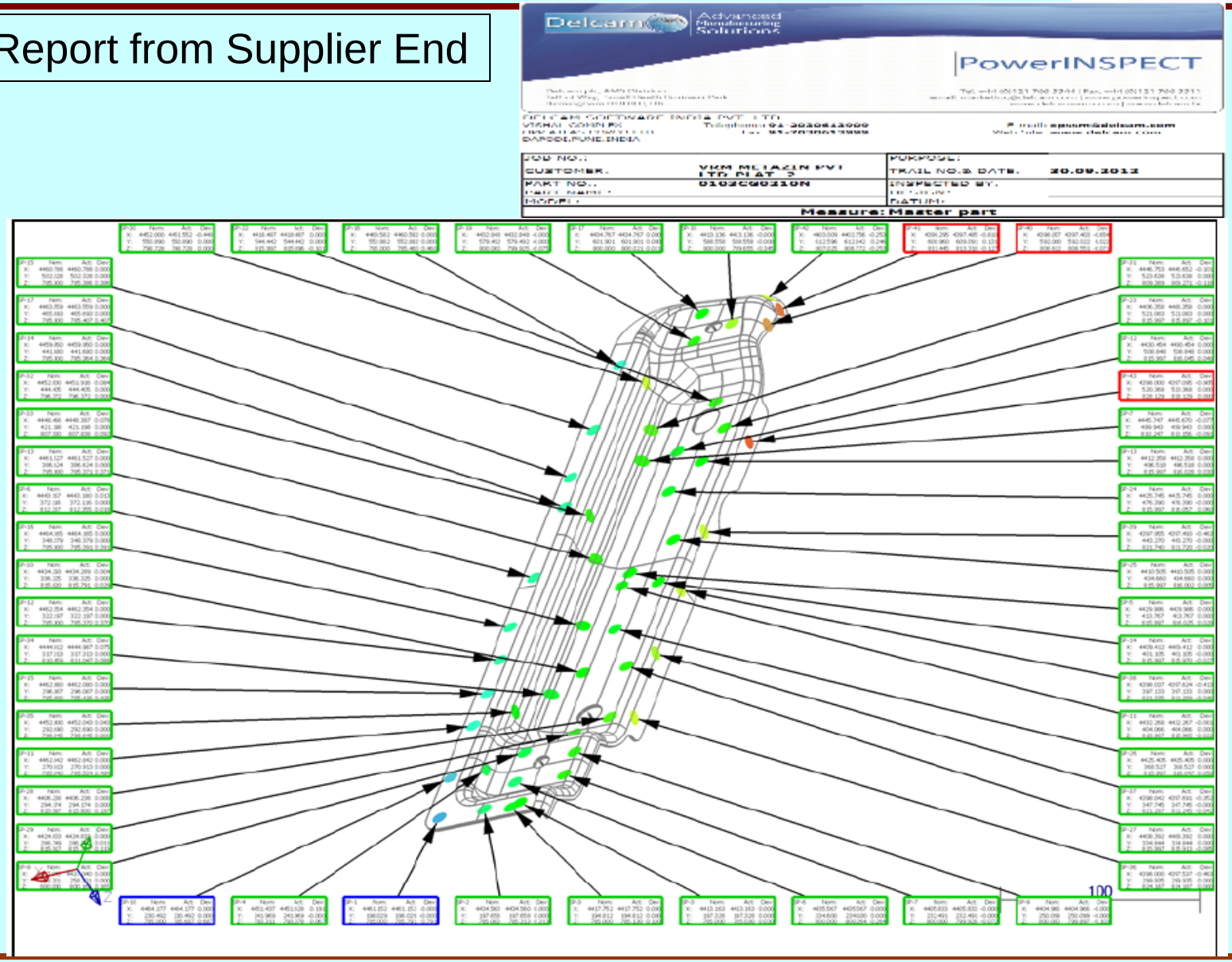
TOOL PREVENTIVE MAINTENANCE SCHEDULE
Plant : VMPL for the year : 2012-2013

#SN	Part No	Operation No & Description		Tool No	Freq	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar
151	0103BG0660N	10	BLANKING	0103BG0660/670N/10	79												
152		20	FORMING	0103BG0660/20	80												
153	0103BG0670N	10	BLANKING	0103BG0660/670N/10	79												
154		20	FORMING	0103BG0670/20	80												
155	0103CG0210N	10	BLANKING	0103CG0210N-10	90												
156		20	FORMING	0103CG0210N-20	270												
157		30	PIERCING	0103CG0210N-30	120												
158	0103EG1190N	10	BLANKING	0103EG1190N/10	90												
159		20	FORMING	0103EG1190N/20	180												
160	0104EL1010N	10	BLANKING	0104EL1010N-10	90												
161		20	FORMING	0104EL1010N-20	90												
162	0105CP0240N/320N	10	BLANKING	0105CP0240N/320N/10	90												
163		20	FORMING I	0105CP0240N/20	180												
164		30	FORMING II	0105CP0320N/20	180												
165		40	PIERCING	0105CP0240N/320N/30	120												
166	0106CAN0090N	10	BLANKING	0106CAN0090N-10	90												
167		20	FORMING	0106CAN0090N-20	180												
168	0112BP0110N	10	BLANKING	0112BP0110N/10	90												
169		20	FORMING I	0112BP0110N/20	210												
170		30	FORMING II	0112BP0110N/30	210												
171		40	PIERCING I	0112BP0110N/40	180												
172		50	PIERCING II	0112BP0110N/50	180												
173	0112BP0120N	10	BLANKING	0112BP0120N/10	90												
174		20	FORMING I	0112BP0120N/20	180												
175		30	FORMING II	0112BP0120N/30	180												

Printed by SACHIN YADAV on 31/10/2012 12:31:28 PM Plan : Actual : Tool PM Schedule

8.1 Updation of documents:

3. CMM Report from Supplier End



8. Standardisation

8.2 Training:

1. OPL

8.3 Additional Audit / Check

1. Supplier Audit
2. 2A-2B audit
3. Supervisory Check sheet

9. Summarise & Plan future actions

9.1 Conclusion:

Back Door inner panel heavy damage & spot hole concern was eliminated by correcting gap in part & panel & part mismatch concern. This shows that importance of part as per specification & Error proofing by Clamp interlocking.

9.2 Horizontal Deployment:

At Three Stages at Scorpio MFL line door assly stages.

9.3 Lessons Learnt:

Things Gone Wrong : Checking & Validating parameters of Weld gun.

Things Gone Right : We challenged the Part specs & with Correction die at supplier end.

Thank You……